

Part 1: Fill-in-the-Blanks (13 * 2 = 26 points)

Award 2 points for each correct answer. Partial credit is generally not applicable for these short answers.

Evolution, Scale & Processes

- **Process by which green plants use sunlight to produce biological molecules is called:**
 - **Answer:** Photosynthesis
- **Phylogenetic trees have "nodes" and "branches"; nodes represent: (Paper 1 variation)**
 - **Answer:** Common ancestors (Acceptable: Speciation events)
- **Phylogenetic trees have "nodes" and "branches"; branches represent: (Paper 2 variation)**
 - **Answer:** Evolutionary lineages (Acceptable: Lines of descent or evolutionary time)
- **Order these items from smallest to largest: (a) animal, (b) atom, (c) cell, (d) disaccharide, (e) protein.**
 - **Answer:** b, d, e, c, a (*Logic: atom < disaccharide < protein < cell < animal*)

Basic Biological Units & Structures

- **The basic unit of life, separated from environment by a membrane is:**
 - **Answer:** Cell
- **The double layered structure formed by fatty acids in water is called a:**
 - **Answer:** Lipid bilayer (Acceptable: Phospholipid bilayer)
- **A single unit of a DNA will consist of the following:**
 - **Answer:** Nucleotide (Acceptable: Deoxynucleotide, or listing the three components: sugar, phosphate, nitrogenous base)

Reactions & Physical Properties

- **Physical property of water resulting in seas with near constant temperature (not hydrogen bonding):**
 - **Answer:** High specific heat capacity (Acceptable: High heat capacity)
- **Bio-polymers are formed by a reaction called: (Paper 1 variation)**
 - **Answer:** Dehydration synthesis (Acceptable: Condensation reaction)
- **Biopolymers are broken down by a reaction called: (Paper 2 variation)**
 - **Answer:** Hydrolysis (Acceptable: Hydrolytic cleavage)

Combinatorics & Sequences

- **For a dna molecule that is 20 base pairs long, how many different strands are possible:**
 - **Answer:** 4^{20}
- **For a protein that is 4 residues long, how many different primary sequences are possible:**
 - **Answer:** 20^4 (Acceptable: 160,000)

Macromolecules, RNA vs DNA & Energy Storage

- **Which nitrogenous base is found in rna but not in dna?**
 - **Answer:** Uracil
- **Which polysaccharide is used for energy storage in plants:**
 - **Answer:** Starch (Acceptable: Amylose or Amylopectin)
- **Which polysaccharide is used for energy storage in animals:**

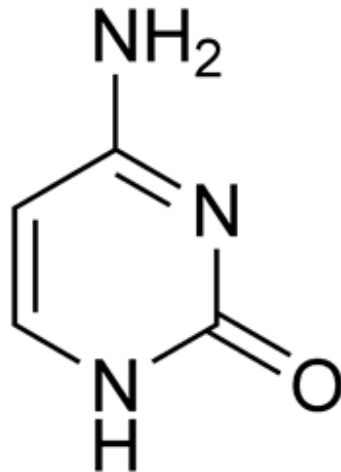
- **Answer:** Glycogen

Part 2: Structural Drawings (Points as indicated)

Nucleobases, Nucleosides, and Nucleotides

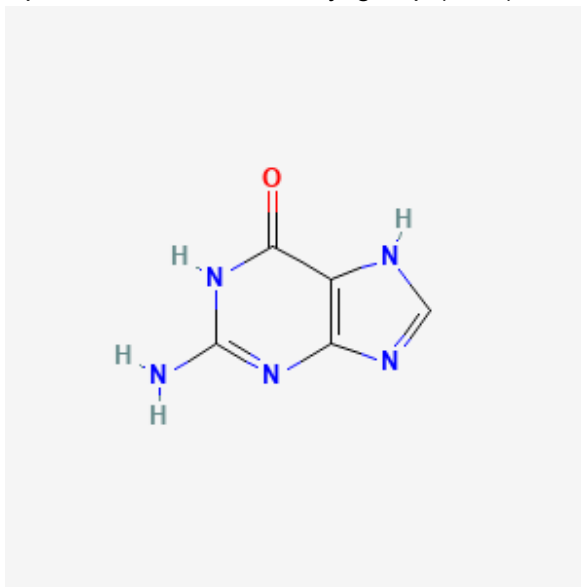
- **Draw the structure of cytosine (2 points)**

- 1 point: Draws a single 6-membered ring (pyrimidine) with nitrogen atoms at positions 1 and 3.
- 1 point: Includes a carbonyl group (C=O) at C2 and an amino group (–NH₂) at C4.



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- **Draw the structure of guanine (2 points) (*Paper 1*)**

- 1 point: Draws a double-ring structure (purine: 6-membered ring fused to a 5-membered ring).
- 1 point: Includes a carbonyl group (C=O) at C6 and an amino group (–NH₂) at C2.

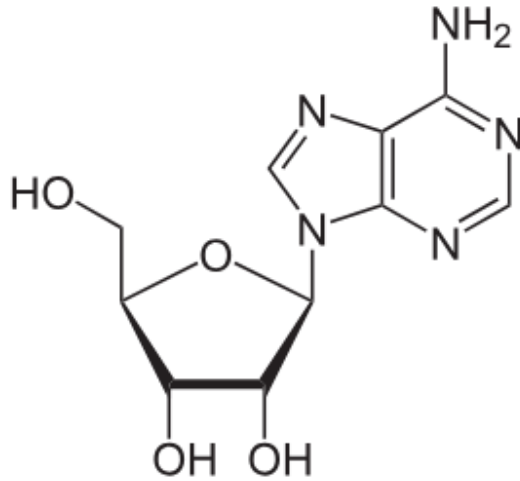


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- **Draw the structure of adenosine (2 points) (*Paper 2*)**

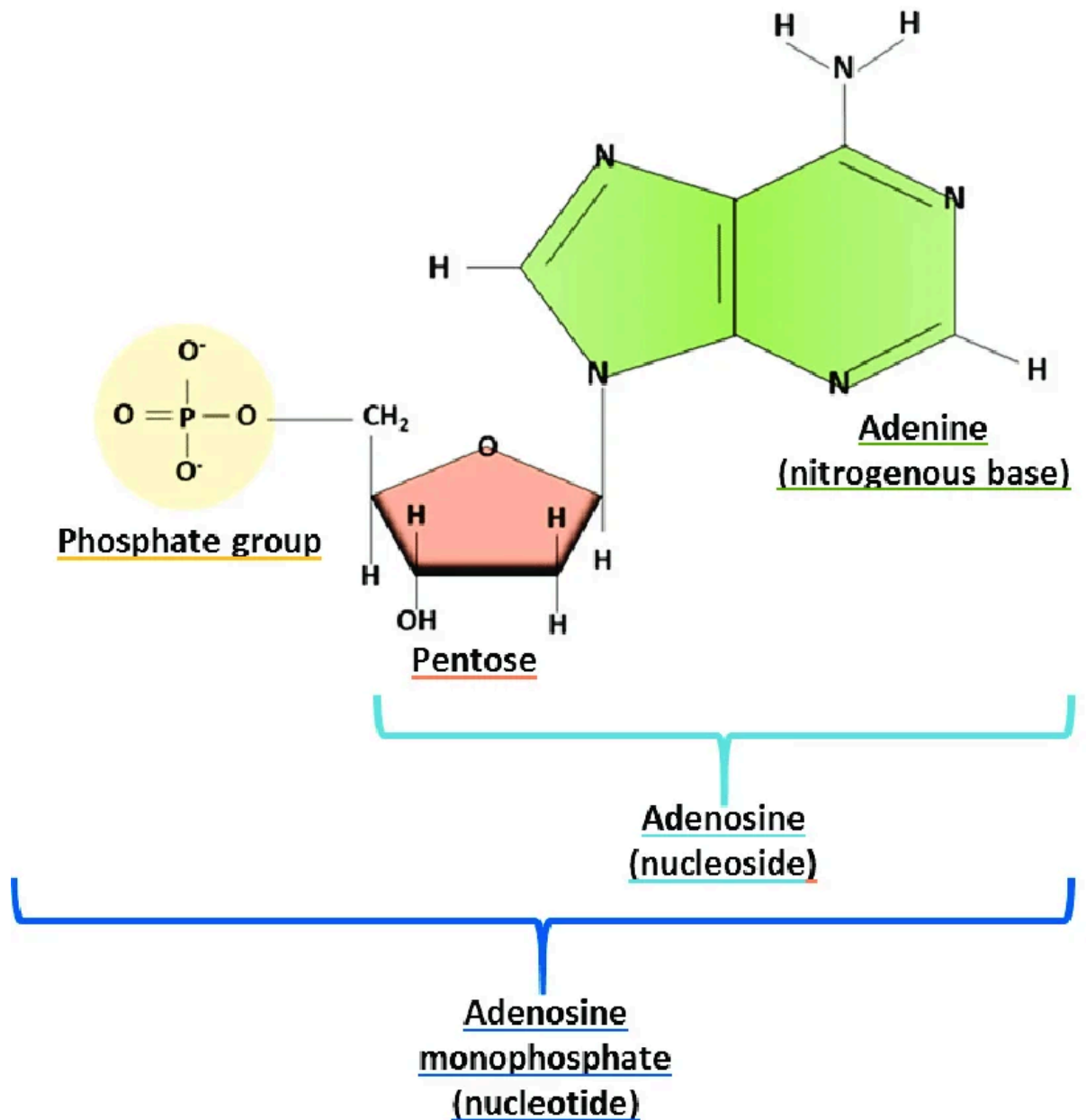
- 1 point: Adenine base is drawn correctly (double-ring purine with an amino group at

C6).

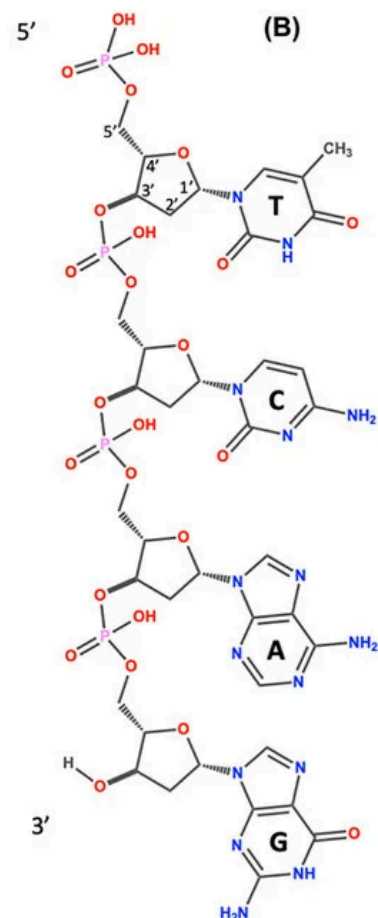
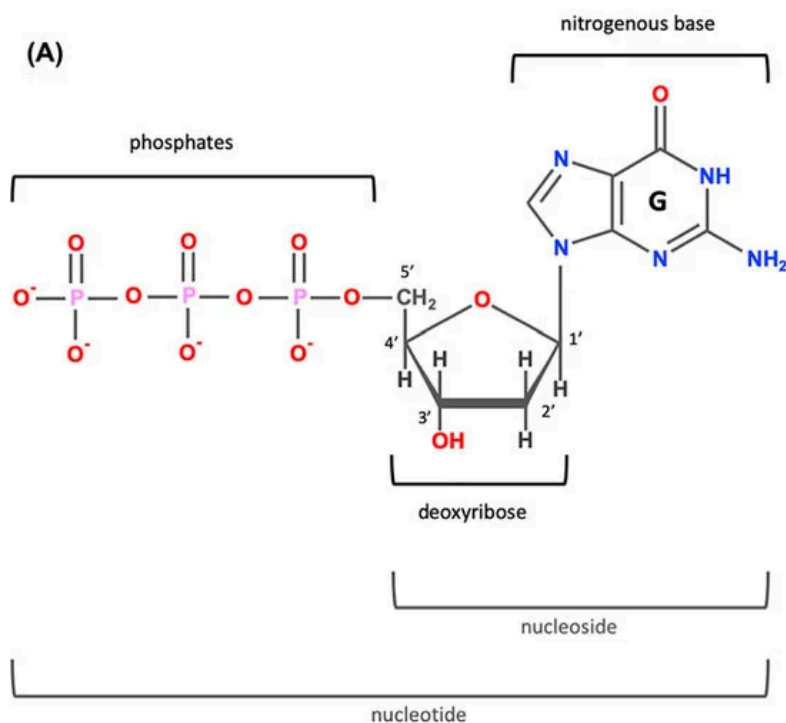
- 1 point: Attached to a 5-membered ribose sugar ring (This is a nucleoside, so ideally no phosphate, but do not penalize heavily if drawn).



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- **Draw the nucleotide of adenosine (3 points)** (*Paper 1*)
 - 1 point: Adenine base drawn correctly.
 - 1 point: Ribose or deoxyribose sugar ring drawn correctly.
 - 1 point: At least one phosphate group (PO₄³⁻) attached to the 5' carbon.

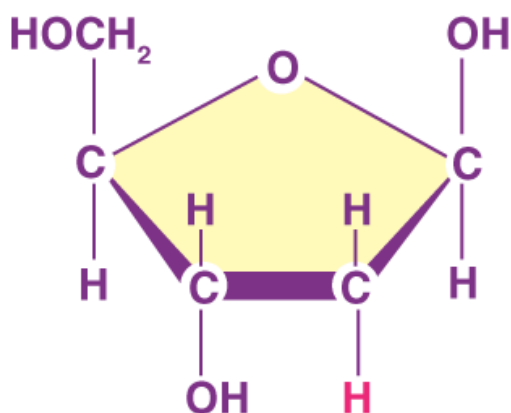


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- **Draw the nucleotide of guanine (3 points) (Paper 2)**
 - 1 point: Guanine base drawn correctly.
 - 1 point: Ribose or deoxyribose sugar ring drawn correctly.
 - 1 point: At least one phosphate group (PO₄³⁻) attached to the 5' carbon.

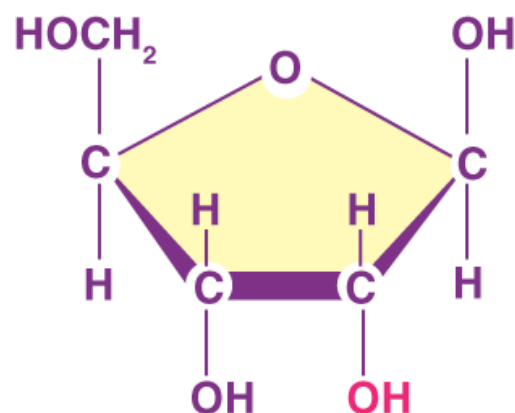


Carbohydrates

- **Name and draw the "3d" structure of a five membered sugar ring seen in biology (2 points) (Paper 1)**
 - 1 point for Name: Ribose or Deoxyribose. More are possible like Fructose, Arabinose or Xylose. Check what the student has drawn if something else is written.
 - 1 point for Drawing: Must draw a 5-membered furanose ring. "3D" requirement means drawing a Haworth projection (flat ring with thickened lower bonds) or envelope conformation.

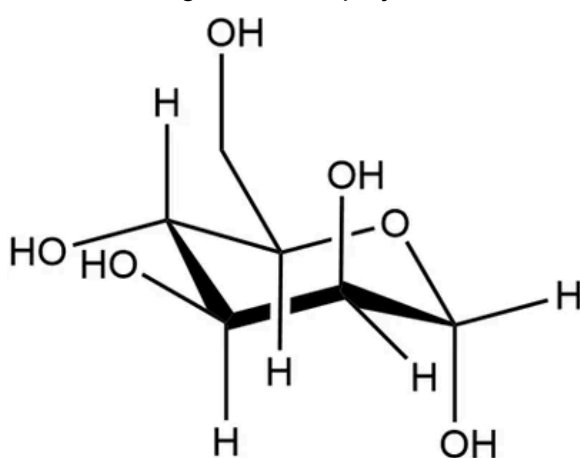


2- Deoxyribose

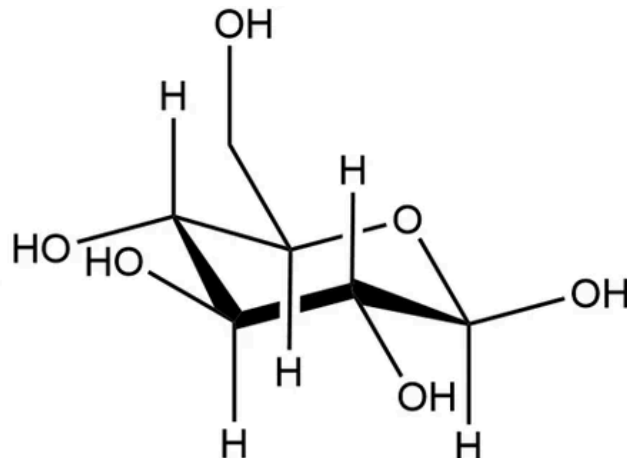


Ribose

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- **Name and draw the "3d" structure of a six membered sugar ring seen in biology (2 points) (Paper 2)**
 - 1 point for Name: Glucose, Galactose, or Mannose. More are possible. Confirm if the student has drawn something else.
 - 1 point for Drawing: Must draw a 6-membered pyranose ring. "3D" requirement means drawing a Haworth projection or a chair conformation.

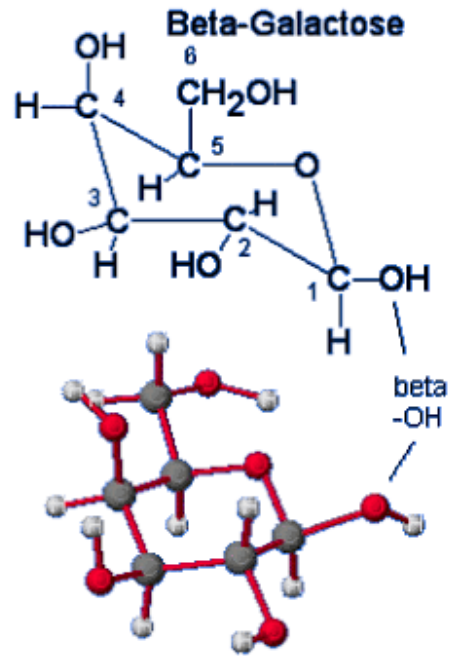


D-Mannose

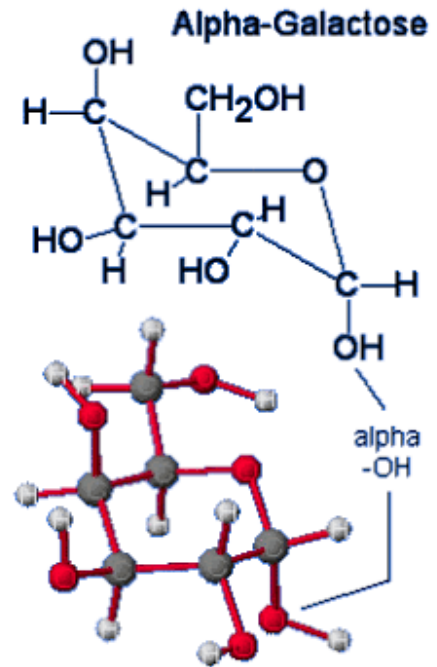


D-Glucose

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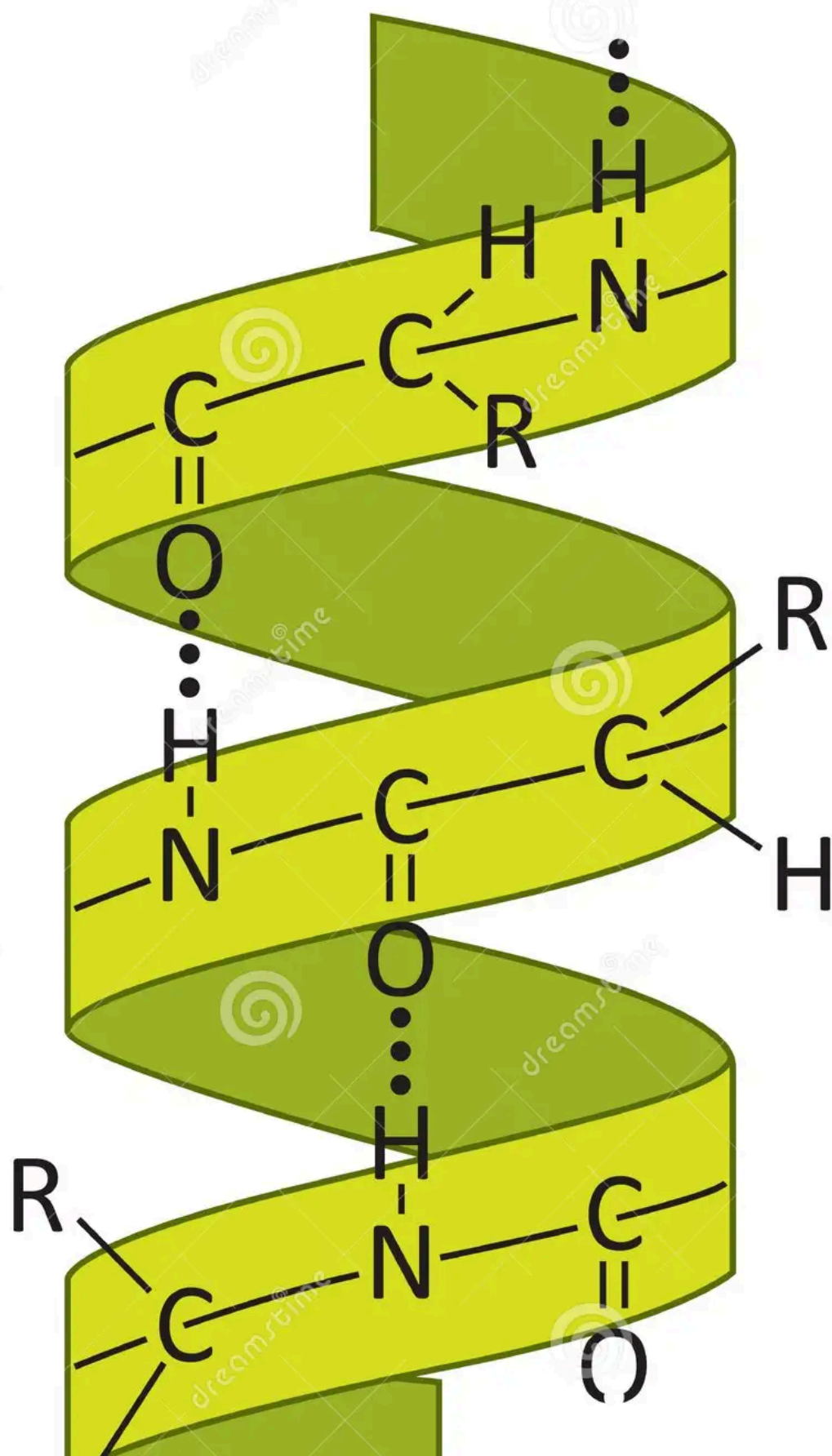


C. Ophardt, c. 2003



Protein Structure

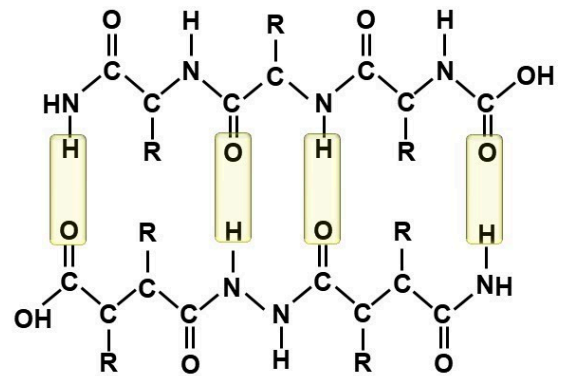
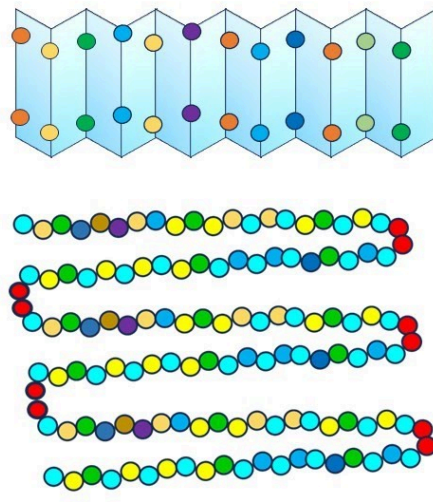
- **Draw the structure of alpha helix (in proteins) showing clearly structural features (3 points) (Paper 1)**
 - 1 point: Shows a continuous right-handed coiled backbone.
 - 1 point: Explicitly draws/labels hydrogen bonds connecting the C=O of one amino acid to the N-H of an amino acid 4 residues down.
 - 1 point: Shows side chains (R-groups) pointing radially outward from the helix.



- **Draw structure of beta sheet (in proteins) showing clearly structural features (3 points) (Paper 2)**
 - 1 point: Shows at least two extended, zig-zagging polypeptide strands alongside each other.
 - 1 point: Explicitly draws/labels hydrogen bonds connecting the C=O group of one strand to the N-H group of the adjacent strand.
 - 1 point: Indicates side chains (R-groups) pointing alternately above and below the sheet plane.

**Quickly
Understand**

β Pleated sheet



Part 3: Short Notes (2 points)

- **Write a short note on requirements for a hypothesis to be scientifically tested (2 points)**
 - Award 1 point for each of the following core concepts (max 2 points):
 1. **Testability:** It must be possible to design a measurable experiment or observation in the natural world.
 2. **Falsifiability:** There must be a way to potentially prove the hypothesis wrong. (e.g., A possible negative answer must exist).